Local–Global Progressive Network for Hyperspectral and Multispectral Image Fusion

Shuyu Zhang, *Member, IEEE*, Yixun Cai, Wangquan He, *Student Member, IEEE*, and Sen Jia, *Senior Member, IEEE*

***Abstract*—Hyperspectral image (HSI) and multispectral image (MSI) fusion aims to generate high-spatial-resolution HSIs (HR-HSIs) by integrating the rich spectral information with the fine spatial details. However, this task remains challenging due to the heterogeneous discrepancies between HSI and MSI in both spatial resolution and spectral dimensionality. The interpolated HSI and MSI often exhibit inconsistent spatial details during feature interaction, complicating reliable cross-modal correspondence and effective exploitation of complementary information. Existing methods struggle to simultaneously suppress cross-modal interference, preserve local details, model global context, and integrate multilevel features. To address these challenges, we propose a local–global progressive network (LGP-Net). First, a spatial–spectral interactive filtering (SSIF) module is designed for early-stage input processing, which enhances discriminative features while suppressing noise and interference caused by dimensional disparities. Second, a local–global collaborative learning (LGCL) module is built to extract stable local details through a physics-guided convolutional branch (PGCB) and cross-modal global dependencies through a multihead cross-shift Transformer (MCST). Finally, a cyclic-refined progressive fusion (CRPF) module is proposed to integrate features through progressively expanding receptive fields and a cyclic accumulation mechanism, facilitating deep multilevel integration while mitigating inconsistencies. Extensive experiments on multiple benchmark datasets demonstrate that LGP-Net significantly outperforms state-of-the-art methods across a range of quantitative metrics. To support reproducibility and future research, we would make the source code publicly available on GitHub.**

***Index Terms*—Cyclic-refined progressive fusion (CRPF), hyperspectral and multispectral images (HSI and MSI) fusion, local–global collaborative learning (LGCL), spatial–spectral interactive filtering (SSIF).**

Received 30 October 2025; revised 22 March 2026 and 11 June 2026; accepted 18 June 2026. Date of publication 30 June 2026; date of current version 16 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U25A20802 and Grant 62271327; in part by Shenzhen Science and Technology Program under Grant JCYJ20240813142308012, Grant RCJC20221008092731042, and Grant KCXFZ20240903093100001; in part by the Project of Department of Education of Guangdong Province under Grant 2023KCXTD029; in part by the Artificial Intelligence for Science (AI4S) Program of Shenzhen University under Grant 2026AI4S11; in part by the Research Team Cultivation Program of Shenzhen University under Grant 2023JCT002; in part by the Open Fund of State Key Laboratory of Remote Sensing and Digital Earth under Grant OFSLRSS202510; in part by the Open Fund of National Earth Observation Data Center under Grant 2025C1; and in part by the Scientific Foundation for Youth Scholars of Shenzhen University under Grant 827-0001080. *(Corresponding author: Sen Jia.)*

The authors are with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: shuyu-zhang@szu.edu.cn; caiyixun2023@email.szu.edu.cn; hewangquan2022@email.szu.edu.cn; senjia@szu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3708339

I. Introduction

HYPERSPECTRAL images (HSIs) are a specialized form of image data that characterize objects by acquiring numerous contiguous spectral bands using hyperspectral cameras. This rich spectral data significantly enhances analytical capabilities, enabling HSIs to achieve superior performance in applications such as environmental monitoring [1], land analysis [2], target tracking [3], and land cover classification [4]. However, hardware limitations typically prevent simultaneous acquisition of both high spatial resolution and high spectral resolution [5]. Consequently, while HSIs provide abundant spectral detail, they often lack fine spatial information. To address this inherent spatial–spectral resolution tradeoff, researchers have proposed fusing the spectral information from low-spatial-resolution HSIs (LR-HSIs) with the spatial information from high-spatial-resolution multispectral images (HR-MSIs). This fusion aims to generate HR-HSIs [6], [7], [8], [9].

In hyperspectral fusion super-resolution (HSR) research, a common assumption is that a mapping relationship exists among the observed HR-HSI, LR-HSI, and HR-MSI [10], [11], [12], [13]. The spatial mapping is typically represented by the point spread function (PSF), while the spectral mapping is captured by the spectral response function (SRF). In some studies, this mapping is predefined as a known degradation model, where the SRF and PSF serve as known priors to regularize the super-resolution solution. These model-driven approaches often employ traditional machine learning techniques. For instance, Yokoya et al. [14] proposed the coupled nonnegative matrix factorization unmixing method. In real-world scenarios, the SRF and PSF are typically unknown and form complex nonlinear relationships that traditional methods struggle to characterize. To enhance applicability and generalization, many researchers have shifted toward blind HSR that regards both functions as unknowns, a substantially more challenging problem.

The advent of deep learning [15], [16] has enabled data-driven methods to demonstrate exceptional capabilities in approximating these complex nonlinear mappings. Recently, convolutional neural networks (CNNs) [17], Transformers [18], generative adversarial networks (GANs) [19], and diffusion models [20] have been widely adopted for blind hyperspectral super-resolution [21], [22], [23]. CNNs excel at extracting local spatial–spectral features. Zhang et al. [24] proposed a spatial–spectral reconstruction network (SSR-Net) for synergistic fusion via a deep CNN. Huang et al. [11]

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

introduced a deep hyperspectral image fusion network (DHIF) with iterative regularization using adaptive 3-D filters. Dong et al. [10] developed a model-guided deep CNN (MoG-DCN), integrating physical degradation and deep-image prior for enhanced interpretability. Transformers leverage attention mechanisms to capture global dependencies. Hu et al. [25] pioneered a Transformer-based fusion network (Fusformer), the first Transformer architecture for HSI fusion. Deng et al. [26] designed a bidirectional dilation Transformer (BDT), which dynamically expands receptive fields via dilation. Ma et al. [13] proposed a dual cross Transformer-based fusion network (DCT) incorporating the cross-attention. In addition, for fusion methods of other network architectures, Xie et al. [27] introduced a 3-D GAN for hyperspectral pansharpening that jointly enhances spatial resolution and preserves spectral fidelity. Yin et al. [28] developed a differential multilevel dense network with adaptive noise adjustment and dynamic diffusion steps. Recently, supervised HSI–MSI fusion methods have advanced toward stronger sequence modeling and adaptive expert learning. Zhu et al. [29] proposed a Mamba-based collaborative implicit neural representation framework for enhanced cross-modal fusion and efficient long-range dependency modeling. He et al. [30] introduced an adaptive expert learning strategy that dynamically selects expert knowledge based on input characteristics to improve fusion flexibility.

Beyond supervised settings, unsupervised HSI–MSI fusion has also made notable progress. Su et al. [31] proposed a dilated transformation-guided unsupervised multimodal learning framework. Li et al. [32] developed an enhanced deep image prior for unsupervised super-resolution, while Gao et al. [33] designed enhanced autoencoders with attention-embedded degradation learning. Additionally, Xu et al. [34] introduced a graph attention-based fusion network, and Zheng et al. [35] proposed an unsupervised pretraining-guided fusion framework. Despite these advances, jointly suppressing cross-modal interference, collaboratively modeling local and global information, and progressively integrating multilevel features in a unified framework remain challenging.

In remote sensing image fusion, hyperspectral and multispectral data exhibit significant dimensional disparities and inherent information redundancy. Effectively extracting their shared and complementary features is critical for enhancing fusion quality. However, current methods face three key limitations. The first issue is about the early-stage degradation. Approaches like BDT and DCT rely on simplistic upsampling strategies during initial fusion, introducing redundancy and interference that destabilize subsequent feature modeling. Early fusion forces the network to reconcile synthetic and real high-frequency components, which might cause representation conflict in shallow layers. While alternative nonearly-stage degradation methods, such as encoder–decoder architectures or parallel streams, can circumvent this specific conflict, they often introduce new challenges. For instance, encoder–decoder models may sacrifice fine spatial details during progressive downsampling, and parallel stream methods face significant computational and memory overhead when extended to high-dimensional hyperspectral data. The second is about contextual modeling deficits. Frameworks such as DHIF and MoG-DCN prioritize local feature extraction but lack cross-modal contextual modeling capabilities, resulting in inadequate structural alignment and underutilization of global information. Local spatial–spectral operations might ignore nonlocal band correlations and cross-modal response relationships. Moreover, the third is about hierarchical integration gaps. Without effective multilevel or cyclic reconstruction mechanisms, the progressive interaction and refinement of features across depths is hindered, reducing cross-scale structural consistency and overall spatial–spectral reconstruction quality.

In this study, we propose a novel local–global progressive network (LGP-Net) for hyperspectral and multispectral image fusion, as shown in Fig. 1. The architecture comprises three hierarchically connected stages. First, a spatial–spectral interactive filtering (SSIF) module is built in the information processing stage. LR-HSI and HR-MSI undergo modality-specific self-guided enhancement, and concurrently, HR-MSI-derived guidance is injected into an HR-HSI prereconstruction module to boost spatial fidelity. Second, a local–global collaborative learning (LGCL) module is established in the feature extraction stage. The global part designs a multihead cross-shift Transformer (MCST) to accumulate intermediate features and perform cross-attention on multisource inputs. The local part implements a physics-guided convolutional branch (PGCB) to preserve structurally aligned local features and enhance fine-grained spectral–spatial consistency. Third, a cyclic-refined progressive fusion (CRPF) module is proposed in the fusion reconstruction stage. Local and global features are integrated through a multiscale convolutional block that progressively expands receptive fields via dilated convolutions. It also cyclically refines feature representations and performs deep cross-scale fusion, iteratively improving spatial–spectral detail reconstruction. Experiments show that the proposed model has superior performance, achieving the best results across multiple benchmark datasets.

In summary, the main contributions are as follows.

1) We design the SSIF for input purification. Unlike existing methods that directly fuse raw inputs, SSIF performs modality-specific self-guided enhancement and cross-modal guided prereconstruction. It jointly suppresses spectral redundancy, spatial inconsistency, and cross-modal interference at the input level, providing clean and well-aligned features for subsequent learning.
2) We propose the LGCL with dual-branch architecture. LGCL introduces a dual-branch design that simultaneously extracts local features via a PGCB and global dependencies via an MCST. This collaborative architecture enables effective integration of fine-grained spatial–spectral details with long-range contextual information.
3) We develop the CRPF for multilevel aggregation. CRPF employs a cyclic refinement mechanism that progressively integrates features across multiple depths, ensuring consistent alignment of shared semantic components. Through iterative updates of the reconstructed HSI and accumulated global representations, it enables deep

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

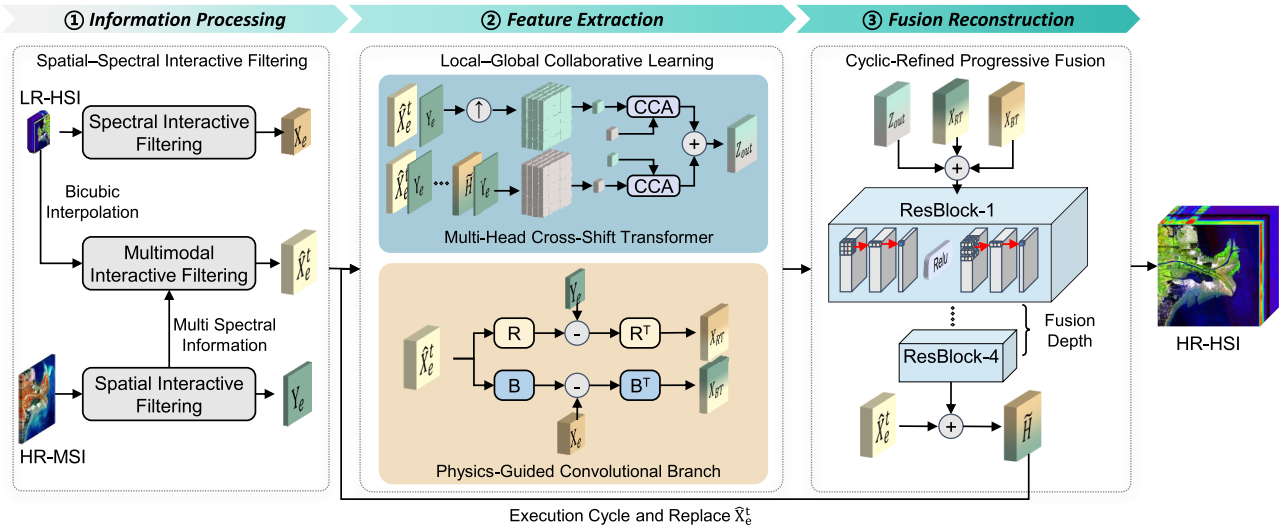


Fig. 1. Overall framework of the proposed LGP-Net for hyperspectral and multispectral image fusion.

multilevel fusion and achieves superior spatial–spectral fidelity.

Beyond module-level design, the key insight of this work is that HSI–MSI fusion should be addressed as a coupled progressive reconstruction problem rather than a single-stage feature mapping. Thus, the contribution of LGP-Net lies not merely in individual modules but in a new architectural paradigm where physics-guided local consistency, global context modeling, and cyclic progressive refinement are jointly optimized. The cyclic accumulation mechanism further serves as an iterative correction strategy that continuously updates both reconstruction state and global representation across cycles, improving cross-scale consistency and reconstruction quality.

II. Related Work

In this section, we provide an overview of research progress in the field of HSI fusion, which also serves as a basis for presenting our subsequent research.

A. CNN-Based Fusion Methods

In recent years, CNNs have demonstrated powerful modeling capabilities in tasks like image super-resolution, object detection, and image generation. By designing networks with varying depths and topologies, CNN-based methods enhance the extraction and fusion of features from both LR-HSI and HR-MSI data. Palsson et al. [36] proposed a 3-D CNN incorporating spatial sampling filters to integrate deep learning into the fusion process. Zhou et al. [37] introduced a pyramid fully convolutional network consisting of an encoder and a pyramid fusion module, which groups spectral bands and learns a nonlinear mapping from multispectral to hyperspectral. Dian et al. [38] presented a three-stage fusion method employing a residual learning network to guide model optimization in subsequent stages. Liu et al. [39] developed a two-stream fusion network (TF-Net) for remote sensing image fusion, which extracts and fuses features from separate LR-HSI and HR-MSI branches before reconstructing the output from a joint representation. Xue et al. [40] introduced a spatial–spectral structured sparse low-rank representation that implicitly models degradation processes and data priors using a deep neural network.

Beyond conventional CNN architectures, emerging deep learning paradigms like GANs and diffusion models have also been introduced into CNN-based fusion methods. For instance, Zhu et al. [22] developed a lightweight adversarial network featuring quadtree implicit sampling, where a discriminator enhances the fidelity of the fused output. Qu et al. [23] proposed a spatial–spectral bilateral cyclic diffusion model that learns the conditional distributions of spatial and spectral super-resolution processes through dual branches, progressively generating high-resolution hyperspectral results. While these CNN-based methods show promise, they face persistent challenges of limited receptive fields that hinder long-range dependency modeling. It degrades the modeling of complex spatial structures and subtle spectral variations, which are crucial for high-fidelity fusion.

B. Transformer-Based Fusion Methods

The success of Transformers in natural language processing has showcased their inherent architectural advantages, leading to widespread adoption across diverse domains and establishing them as fundamental deep learning tools comparable to CNNs. Hu et al. [25] were among the first to apply the Transformer architecture to HSR. They proposed Fusformer, leveraging a global attention mechanism to capture long-range dependencies, thereby opening a new research direction in the field. Subsequently, Ma et al. [41] introduced the 3D-CNN and Transformer prior network (3DT), a fusion network that effectively integrates Transformers and 3-D CNNs to learn powerful hyperspectral priors for enhanced fusion. Further advancing this approach, Deng et al. [42] developed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

the pyramid shuffle-and-reshuffle Transformer (PSRT), which combines shuffle-and-reshuffle mechanisms with multiscale feature extraction to model both local and global representations.

To enhance model performance and generalization, Jia et al. [43] designed the multiscale spatial–spectral Transformer, incorporating a self-supervised pretraining strategy to stabilize training and improve adaptability. Wang et al. [44] introduced the multihierarchical cross Transformer, which employs cross-attention mechanisms to extract complementary features from LR-HSI and HR-MSI inputs. Building on cross-modal integration, Ma et al. [13] proposed the directional paired cross-modal Transformer, utilizing directional paired multihead cross-attention to better capture mutual information between modalities. Concurrently, He et al. [45] further advanced multiscale modeling with their local–global collaborative Transformer, constructing deep multiscale representations of HR-MSI and LR-HSI features. While Transformer-based models significantly improve feature modeling capabilities and provide a flexible framework for multisource data fusion via attention mechanisms, they often suffer from high model complexity and substantial computational costs.

III. Methodology

We propose an LGP-Net method for hyperspectral and multispectral image fusion, which achieves super-resolved reconstruction of HR-HSIs, as shown in Fig. 1.

A. Model Framework

The LGP-Net is a dual-input model that processes an LR-HSI $\mathbf{X} \in \mathbb{R}^{h\times w\times B}$ and an HR multispectral image $\mathbf{Y} \in \mathbb{R}^{H\times W\times b}$. $h \times w \times B$ represent the height, width, and band dimensions of LR-HSI, respectively, and $H \times W \times b$ represent the height, width, and band dimensions of HR-MSI, respectively. These inputs undergo preprocessing through three dedicated submodules within the SSIF module, operating on the LR-HSI $\mathbf{X}$, the HR-MSI $\mathbf{Y}$, and an upsampled HSI $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times B}$ generated by applying bicubic interpolation to $\mathbf{X}$. LR-HSI and HR-MSI data undergo layer-wise decomposition, enabling fine-grained spatial–spectral feature disentanglement. Each modality independently recovers a preliminary target image using modality-specific guidance. The upsampled HSI is decomposed to extract hierarchical spectral guidance, which fuses with spatial guidance from HR-MSI's self-processed output, enabling the reconstruction of a spatially and spectrally consistent upsampled HSI. The SSIF module outputs preprocessed and prereconstructed basis matrices of $\mathbf{X}_e \in \mathbb{R}^{h\times w\times B}$, $\mathbf{Y}_e \in \mathbb{R}^{H\times W\times b}$, and $\hat{\mathbf{X}}_e \in \mathbb{R}^{H\times W\times B}$.

Following preprocessing and preliminary reconstruction in the SSIF module, $\mathbf{X}_e$, $\mathbf{Y}_e$, and $\hat{\mathbf{X}}_e$ are then used for local–global feature extraction in the LGCL module. Local learning is performed through the PGCB, and global learning is implemented by the MCST. The PGCB leverages physical priors to extract stable and locally aligned features, which excel at preserving fine-grained spectral–spatial consistency. $\hat{\mathbf{X}}_e$ and $\mathbf{Y}_e$ are fed into R branch for SRF simulation, yielding the output $\mathbf{X}_{\mathrm{RT}}$. $\hat{\mathbf{X}}_e$ and $\mathbf{X}_e$ are fed into the B branch for PSF simulation, resulting in the output $\mathbf{X}_{\mathrm{BT}}$. Collaboratively, the MCST accumulates intermediate feature maps across stages to preserve multiscale semantic cues, and its core design of cross-convolutional attention (CCA) is a bidirectional mechanism where local features modulate global attention. $\hat{\mathbf{X}}_e$ and $\mathbf{Y}_e$ are concatenated to form a multisource feature matrix $\mathbf{Z}^1$. The multisource feature matrices from each iteration are then concatenated to form an accumulated feature matrix $\mathbf{Z}'$. Both $\mathbf{Z}^1$ and $\mathbf{Z}'$ are adjusted and fed into two CCA branches, performing attention calculation sequentially as primary and auxiliary information. The outputs of two CCA branches are summed to obtain the MCST output $\mathbf{Z}_{\mathrm{out}}$.

The local features $\mathbf{X}_{\mathrm{RT}}$ and $\mathbf{X}_{\mathrm{BT}}$ from the PGCB, along with the global features $\mathbf{Z}_{\mathrm{out}}$ from the MCST, are integrated into the CRPF module for fusion reconstruction. These three feature matrices undergo weighted summation using two adaptive parameters to produce a unified feature matrix $\mathbf{H}$. $\mathbf{H}$ is then processed through a multilevel fusion module, where convolutional blocks with progressively increasing receptive fields perform hierarchical feature fusion to obtain $\mathbf{H}'$. Then, residual learning is implemented by adding the multilevel fusion output $\mathbf{H}'$ to the basis feature $\hat{\mathbf{X}}_e$, which generates an intermediate output $\tilde{\mathbf{H}}$ that initiates the next cycle. The model executes multiple iterative cycles, with only the first cycle processing inputs through the SSIF module. In subsequent cycles ($t \geq 2$), the CRPF output $\tilde{\mathbf{H}}$ directly replaces $\hat{\mathbf{X}}_e$ as input, denoted $\hat{\mathbf{X}}_e^t$, where t is the cycle index. Specifically, in the first cycle, $\hat{\mathbf{X}}_e^1$ equals to the SSIF output $\hat{\mathbf{X}}_e$, and in the other cycles, $\hat{\mathbf{X}}_e^t$ corresponds to the previous cycle's CRPF output $\tilde{\mathbf{H}}$. Through cyclic feature refinement, representations are iteratively updated via feedback connections, ensuring consistent alignment of shared semantic components across refinement cycles. After multiple cycles, the final HR-HSI $\mathbf{H}_{\mathrm{out}}$ is reconstructed.

B. Spatial–Spectral Interactive Filtering

The SSIF module comprises three specialized submodules that perform hyperspectral preprocessing on LR-HSI $\mathbf{X}$, multispectral preprocessing on HR-MSI $\mathbf{Y}$, and hyperspectral prereconstruction on interpolated HR-HSI $\hat{\mathbf{X}}$ obtained from $\mathbf{X}$, respectively. Crucially, $\hat{\mathbf{X}}$ contains no additional information beyond its parent $\mathbf{X}$ since it is generated directly through bicubic interpolation, creating substantial redundant information. Furthermore, mutual interference exists between $\mathbf{X}$ and $\mathbf{Y}$ due to inconsistent spectral representations of the same scene across different band resolutions. To address these issues, we design two dedicated filtering submodules: the spectral interactive filtering (SpeIF) submodule to align the spectral representations between modalities, and the spatial interactive filtering (SpaIF) submodule to compress the spatial redundancy between $\mathbf{X}$ and $\hat{\mathbf{X}}$. However, even after these separate corrections, residual cross-modal interference still persists, stemming from the complex interactions between the spatially and spectrally corrected features. Therefore, the multimodal interactive filtering (MIF) submodule is introduced to holistically refine the features and eliminate this

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

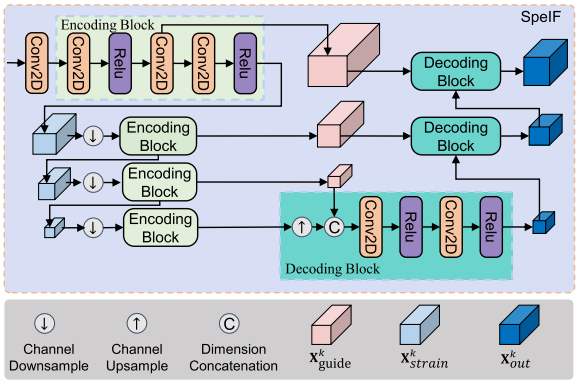


Fig. 2. Structure of SpeIF submodule.

remaining interference. Each submodule implements a dual-process architecture consisting of decomposition with feature representation and reconstruction with feature guidance.

For example, in the SpeIF submodule (Fig. 2), the decomposition process is given by

$$\mathbf{X}^k_{\text{encode_in}} = \begin{cases} \text{Conv}\left(\mathbf{X}\right), & k = 1 \\ \text{Conv}\left(\mathbf{X}^{k-1}_{\text{strain}} \downarrow_{\text{cha}}\right), & k = 2,3,4 \end{cases} \tag{1}$$

$$\mathbf{X}^k_{\text{guide}} = \text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{X}^k_{\text{encode_in}}\right)\right)\right) \tag{2}$$

$$\mathbf{X}^k_{\text{strain}} = \text{ReLU}\left(\text{Conv}\left(\mathbf{X}^k_{\text{guide}} \uparrow_{\text{cha}}\right)\right). \tag{3}$$

In (1), $\mathbf{X}$ represents the LR-HSI input, and $\mathbf{X}^k_{\text{encode_in}}$ means the input feature of encoding block. Here, k denotes the block number. The initial convolution at $k = 1$ aligns spatial–spectral dimensions while preserving the channel count of $\mathbf{X}$. Subsequent blocks ($k \geq 2$) apply the Conv($\downarrow_{\text{cha}}$) operation to perform block-wise channel downsampling until matching the dimension of $\mathbf{Y}$. The channel operations in SpeIF serve to progressively compress and subsequently restore spectral dimensionality. Equation (2) generates guidance matrix $\mathbf{X}^k_{\text{guide}}$ through a two-stage convolution with rectified linear unit (ReLU) activation, preparing essential features for downstream reconstruction. Then, (3) performs channel upsampling Conv($\uparrow_{\text{cha}}$) and ReLU on the guidance matrix to obtain propagation feature $\mathbf{X}^k_{\text{strain}}$ for the subsequent encoding block. Similarly, the decomposition process of the SpaIF submodule is as follows:

$$\mathbf{Y}^k_{\text{encode_in}} = \begin{cases} \text{Conv}\left(\mathbf{Y}\right), & k = 1 \\ \text{Conv}\left(\mathbf{Y}^{k-1}_{\text{strain}} \downarrow_{\text{spa}}\right), & k = 2,3,4 \end{cases} \tag{4}$$

$$\mathbf{Y}^k_{\text{guide}} = \text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{Y}^k_{\text{encode_in}}\right)\right)\right) \tag{5}$$

$$\mathbf{Y}^k_{\text{strain}} = \text{ReLU}\left(\text{Conv}\left(\mathbf{Y}^k_{\text{guide}} \uparrow_{\text{cha}}\right)\right). \tag{6}$$

In (4), $\mathbf{Y}$ represents the HR-MSI input, and $\mathbf{Y}^k_{\text{encode_in}}$ means the encoding block input. Different from SpeIF, the SpaIF applies the Conv($\downarrow_{\text{spa}}$) operation to perform spatial downsampling when $k \geq 2$. The spatial operations in SpaIF and MIF are designed to construct a multiscale spatial hierarchy. Each downsampling operation halves the spatial resolution, so three successive reductions yield an $8\times$ smaller spatial dimension. Guidance matrix $\mathbf{Y}^k_{\text{guide}}$ and propagation feature $\mathbf{Y}^k_{\text{strain}}$ are generated by (5) and (6), respectively. Considering that $\hat{\mathbf{X}}$ and $\mathbf{Y}$ have the same spatial dimensions, the same calculation process is performed for $\hat{\mathbf{X}}$ in the MIF submodule, and guidance matrix $\hat{\mathbf{X}}^k_{\text{guide}}$ and propagation feature $\hat{\mathbf{X}}^k_{\text{strain}}$ are obtained.

For the reconstruction process, it is designed symmetrically to the decomposition process, utilizing encoded features and guidance matrices to restore the original size. The reconstruction process of the SpeIF submodule is given by

$$\mathbf{X}^k_{\text{up}} = \begin{cases} \text{Conv}\left(\mathbf{X}^{k+1}_{\text{strain}} \uparrow_{\text{cha}}\right), & k = 3 \\ \text{Conv}\left(\mathbf{X}^{k+1}_{\text{out}} \uparrow_{\text{cha}}\right), & k = 1,2 \end{cases} \tag{7}$$

$$\mathbf{X}^k_{\text{cat}} = \text{Concat}\left(\mathbf{X}^k_{\text{up}}, \mathbf{X}^k_{\text{guide}}\right) \tag{8}$$

$$\mathbf{X}^k_{\text{out}} = \text{ReLU}\left(\text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{X}^k_{\text{cat}} \downarrow_{\text{cha}}\right)\right)\right)\right). \tag{9}$$

In (7) and (8), the decomposed feature $\mathbf{X}^{k+1}_{\text{strain}}$ or reconstructed feature $\mathbf{X}^{k+1}_{\text{out}}$ undergoes block-wise channel upsampling Conv($\uparrow_{\text{cha}}$) to align upsampled feature $\mathbf{X}^k_{\text{up}}$ with guidance matrix $\mathbf{X}^k_{\text{guide}}$ in spectral dimension for concatenation. In (9), the concatenated feature $\mathbf{X}^k_{\text{cat}}$ undergoes channel downsampling Conv($\downarrow_{\text{cha}}$) and convolution with activation to generate reconstructed feature $\mathbf{X}^k_{\text{out}}$, which is propagated to the preceding decoding block. Similarly, the reconstruction process of the SpaIF submodule is as follows:

$$\mathbf{Y}^k_{\text{up}} = \begin{cases} \text{Conv}\left(\mathbf{Y}^{k+1}_{\text{strain}} \uparrow_{\text{spa}}\right), & k = 3 \\ \text{Conv}\left(\mathbf{Y}^{k+1}_{\text{out}} \uparrow_{\text{spa}}\right), & k = 1,2 \end{cases} \tag{10}$$

$$\mathbf{Y}^k_{\text{cat}} = \text{Concat}\left(\mathbf{Y}^k_{\text{up}}, \mathbf{Y}^k_{\text{guide}}\right) \tag{11}$$

$$\mathbf{Y}^k_{\text{out}} = \text{ReLU}\left(\text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{Y}^k_{\text{cat}} \downarrow_{\text{cha}}\right)\right)\right)\right). \tag{12}$$

In (10) and (11), the decomposed feature $\mathbf{Y}^{k+1}_{\text{strain}}$ or reconstructed feature $\mathbf{Y}^{k+1}_{\text{out}}$ undergoes block-wise spatial upsampling Conv($\uparrow_{\text{spa}}$) to align $\mathbf{Y}^k_{\text{up}}$ with $\mathbf{Y}^k_{\text{guide}}$ in spatial dimension for concatenation. Reconstructed feature $\mathbf{Y}^k_{\text{out}}$ is produced through (12) for propagation. The reconstruction process of the MIF submodule is similar to the SpaIF, but the MIF ensures that both hyperspectral and multispectral information participate in reconstruction guidance, enhancing the expressive ability of interpolated HSI $\hat{\mathbf{X}}$ for subsequent learning and fusion. It is implemented differently through the feature concatenation, which is represented by

$$\hat{\mathbf{X}}^k_{\text{cat}} = \text{Concat}\left(\hat{\mathbf{X}}^k_{\text{up}}, \hat{\mathbf{X}}^k_{\text{guide}}, \mathbf{Y}^k_{\text{guide}}\right). \tag{13}$$

After the reconstruction process of SpeIF, SpaIF, and MIF submodules, the obtained $\mathbf{X}_e$, $\mathbf{Y}_e$, and $\hat{\mathbf{X}}_e$ have the characteristics of spatial–spectral filtering for downstream analysis, reducing the impact of redundancy and interference.

C. Local–Global Collaborative Learning

The LGCL integrates two interactive branches: PGCB extracts stable local features, and MCST models complex global dependencies. In the PGCB, the SSIF-derived matrices $\mathbf{X}_e$, $\mathbf{Y}_e$, and $\hat{\mathbf{X}}_e$ serve as inputs. Let $\hat{\mathbf{X}}^t_e$ denote the fused feature during tth cycle. $\hat{\mathbf{X}}^t_e$ is first applied spatial and spectral adjustments using convolution, after which differences are computed relative to $\mathbf{X}_e$ and $\mathbf{Y}_e$, respectively. Then, transposed convolution is utilized to restore original dual-resolution dimensions, obtaining locally detailed features. $\hat{\mathbf{X}}^t_e$ and $\mathbf{Y}_e$ are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

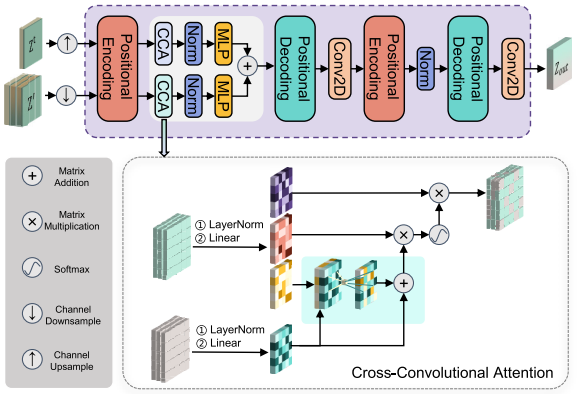


Fig. 3. Structure of MCST.

fed into R branch for SRF simulation, yielding the output $\mathbf{X}_{\mathrm{RT}}$. $\hat{\mathbf{X}}_e^t$ and $\mathbf{X}_e$ are fed into the B branch for PSF simulation, resulting in the output $\mathbf{X}_{\mathrm{BT}}$. The operations are defined as follows:

$$\mathbf{X}_{\mathrm{B}} = \mathrm{Conv}\left(\hat{\mathbf{X}}_e^t \downarrow_{\mathrm{spa}}\right) \tag{14}$$

$$\mathbf{X}_{\mathrm{R}} = \mathrm{Conv}\left(\hat{\mathbf{X}}_e^t \downarrow_{\mathrm{cha}}\right) \tag{15}$$

$$\mathbf{X}_{\mathrm{BT}}^t = \mathrm{Transposed_Conv}\left(\left(\mathbf{X}_{\mathrm{B}} - \mathbf{X}_e\right) \uparrow_{\mathrm{spa}}\right) \tag{16}$$

$$\mathbf{X}_{\mathrm{RT}}^t = \mathrm{Transposed_Conv}\left(\left(\mathbf{X}_{\mathrm{R}} - \mathbf{Y}_e\right) \uparrow_{\mathrm{cha}}\right). \tag{17}$$

In (14) and (15), $\mathrm{Conv}(\downarrow_{\mathrm{spa}})$ and $\mathrm{Conv}(\downarrow_{\mathrm{cha}})$ are employed to perform spatial and channel downsampling on $\hat{\mathbf{X}}_e^t$, respectively. Then, $\mathrm{Transposed_Conv}(\uparrow_{\mathrm{spa}})$ performs spatial resolution recovery using transposed convolution on the difference of $\mathbf{X}_{\mathrm{B}}$ and $\mathbf{X}_e$, and $\mathrm{Transposed_Conv}(\uparrow_{\mathrm{cha}})$ performs spectral recovery on the difference of $\mathbf{X}_{\mathrm{R}}$ and $\mathbf{Y}_e$. These operations are learnable transformations that enforce cross-domain consistency and convert discrepancies into informative local features. Equations (16) and (17) yield the spatial feature matrix $\mathbf{X}_{\mathrm{BT}}^t$ and the spectral feature matrix $\mathbf{X}_{\mathrm{RT}}^t$, respectively. Both feature matrices are subsequently processed in the CRPF module. The index t denotes cycle progression, where for $t \geq 2$, the previous cycle's output initializes $\hat{\mathbf{X}}_e^t$ for multisource feature matrix generation.

The MCST module extracts global spectral–spatial features through CCA, multilayer perceptron (MLP), and 2-D convolution (Conv2D), as shown in Fig. 3. For one input of concatenated feature learning, in the first cycle, MIF-derived $\hat{\mathbf{X}}_e$ is concatenated with $\mathbf{Y}_e$ to form the multisource matrix $\mathbf{Z}^1$. In each subsequent cycle, $\mathbf{Z}^t$ is refreshed by aggregating the previous cycle's output $\tilde{\mathbf{H}}$ with $\mathbf{Y}_e$, maintaining spatial–spectral consistency through dimensionality adjustment. For the other input of accumulated feature handling, in the first cycle, $\mathbf{Z}'$ is generated by applying dimension-adjusted convolution to $\mathbf{Z}^1$. In the following cycles ($t \geq 2$), current $\mathbf{Z}^t$ is concatenated into accumulation matrix $\mathbf{Z}_{\mathrm{ac}}$, then applied spectral-dimension-adjusting convolution to obtain $\mathbf{Z}'$. The MCST processes both $\mathbf{Z}^t$ and $\mathbf{Z}'$ to extract global features, and dimensionality adjustment is given by

$$\mathbf{Z}^t = \mathrm{Conv}\left(\mathrm{Concat}\left(\hat{\mathbf{X}}_e^t, \mathbf{Y_e}\right) \uparrow_{\mathrm{cha}}\right) \tag{18}$$

$$\mathbf{Z}_{\mathrm{ac}} = \mathrm{Concat}\left(\mathbf{Z}_{\mathrm{ac}}, \mathbf{Z}^t\right) \tag{19}$$

$$\mathbf{Z}' = \mathrm{Conv}\left(\mathbf{Z}_{\mathrm{ac}} \downarrow_{\mathrm{cha}}\right). \tag{20}$$

After $\mathbf{Z}^t$ and $\mathbf{Z}'$ are input to MCST, they undergo the positional embedding encoder, and global features $\mathbf{Z}_f$ are extracted using CCA. This module employs a dual-branch CCA to capture complex spatial–spectral interdependencies. Through input role-swapping, it generates complementary feature representations: $\mathbf{Z}_{\text{ms-ac}}$ regards multisource feature as primary and accumulated feature as secondary, while $\mathbf{Z}_{\text{ac-ms}}$ exchanges primary and secondary features. The role-swapping mechanism ensures comprehensive cross-modal interaction by alternating feature dominance in each branch. The implementations are formalized by

$$\mathbf{Z}_{\text{ms-ac}} = \mathrm{MLP}\left(\mathrm{Norm}\left(\mathrm{CCA}\left(\mathbf{Z}^t, \mathbf{Z}'\right)\right)\right) \tag{21}$$

$$\mathbf{Z}_{\text{ac-ms}} = \mathrm{MLP}\left(\mathrm{Norm}\left(\mathrm{CCA}\left(\mathbf{Z}', \mathbf{Z}^t\right)\right)\right) \tag{22}$$

where $\mathrm{Norm}(\cdot)$ means the normalization operation. The detailed implementation of the CCA is as follows. Key $\mathbf{K}$, query $\mathbf{Q}$, and value $\mathbf{V}$ matrices are obtained from $\mathbf{Z}^t$ through linear transformation. Then, the cross-convolutional key is generated by

$$\mathbf{K}' = \mathbf{K} + \mathrm{Conv}\left(\mathbf{K} + \mathrm{Linear}\left(\mathrm{Norm}\left(\mathbf{Z}'\right)\right)\right). \tag{23}$$

In (23), the auxiliary key matrix derived from $\mathbf{Z}'$ is first added to $\mathbf{K}$, and a subsequent convolution is applied. In this way, neighboring pixel representations can modulate the key embedding of the central pixel. CCA is a cross-attention-based fusion operator, which selectively injects complementary information from one feature branch into another during progressive reconstruction. In addition, convolutional projections are applied before attention computation to preserve local inductive bias and adapt embeddings to the spatial–spectral fusion task. This enables effective integration of global information with local features in subsequent reconstruction.

Following positional encoding, both $\mathbf{Z}^t$ and $\mathbf{Z}'$ are partitioned into local windows. $\mathbf{Z}^t$ is divided into m nonoverlapping window patches $\{\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_m\}$, and $\mathbf{Z}'$ is split into corresponding patches $\{\mathbf{P}'_1, \mathbf{P}'_2, \ldots, \mathbf{P}'_m\}$, where each $\mathbf{P}_i, \mathbf{P}'_i \in \mathbb{R}^{L^2 \times f}$ for $i \in [1, m]$ with $m = H \times W / L^2$. Here, L denotes the window side length and f denotes the channel dimension. These patches undergo layer normalization followed by linear projection. Using $\mathbf{Z}^t$ patches as an example, the projection is computed by

$$\mathbf{Q}_i = \mathbf{P}_i \mathbf{W}^Q, \quad \mathbf{K}_i = \mathbf{P}_i \mathbf{W}^K, \quad \mathbf{V}_i = \mathbf{P}_i \mathbf{W}^V \tag{24}$$

where $\mathbf{W}^Q$, $\mathbf{W}^K$, and $\mathbf{W}^V \in \mathbb{R}^{f \times d}$ are projection matrices, and attention is computed in a multihead parallel manner as follows:

$$\mathrm{Attention} = \mathrm{Softmax}\left(\frac{\mathbf{Q}\left(\mathbf{K}'\right)^T}{\sqrt{d}}\right) \tag{25}$$

$$\mathbf{Z}^0_{\text{ms-ac}} = \mathrm{Linear}\left(\mathrm{Attention} \cdot \mathbf{V}\right) \tag{26}$$

$$\mathbf{Z}_{\text{ms-ac}} = \mathrm{MLP}\left(\mathrm{Norm}\left(\mathbf{Z}^0_{\text{ms-ac}}\right)\right) \tag{27}$$

where $\mathbf{Q}$ and $\mathbf{V}$ are derived exclusively from the multisource matrix $\mathbf{Z}^t$, while $\mathbf{K}'$ is generated jointly from $\mathbf{Z}^t$ and accumulated feature matrix $\mathbf{Z}'$. The CCA computes an attention map through $\mathbf{K}' \times \mathbf{Q}$, applies the attention map to $\mathbf{V}$ via matrix

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 LGP-Net for HSI and MSI Fusion

Input: LR-HSI: $\mathbf{X} \in \mathbb{R}^{h\times w\times B}$, HR-MSI: $\mathbf{Y} \in \mathbb{R}^{H\times W\times b}$
Output: reconstructed HR-HSI: $\mathbf{H}_{\text{out}} \in \mathbb{R}^{H\times W\times B}$

Begin
perform bicubic interpolation on the LR-HSI to obtain upsampled representation $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times B}$
obtain preprocessed features $\mathbf{X}_e \in \mathbb{R}^{h\times w\times B}$, $\mathbf{Y}_e \in \mathbb{R}^{H\times W\times b}$, and $\hat{\mathbf{X}}_e \in \mathbb{R}^{H\times W\times B}$ through SSIF with (1)-(13), $\hat{\mathbf{X}}_\mathbf{e}^\mathbf{1} = \hat{\mathbf{X}}_e$
concatenate $\hat{\mathbf{X}}_\mathbf{e}^\mathbf{1}$ and $\mathbf{Y}_e$ along channel to obtain $\mathbf{Z}^1$
for each cycle t **do**
  spatial and channel dimension reduction with (14)-(15)
  obtain local correction features $\mathbf{X}_{\text{BT}}^t$ and $\mathbf{X}_{\text{RT}}^t$ through PGCB with (16)-(17)
  channel concatenation and adjustment with (18)-(20)
  update global accumulated feature $\mathbf{Z}_{\text{ac}}^t$ using current-cycle input and $\mathbf{Z}_{\text{ac}}^{t-1}$
  obtain global feature $\mathbf{Z}_{\text{out}}$ through MCST with (21)-(29)
  obtain reconstructed feature $\tilde{\mathbf{H}}$ through CRPF with (30)-(33)
  update $\hat{\mathbf{X}}_e^{t+1} = \tilde{\mathbf{H}}$
  update $\mathbf{Z}^{t+1}$ with (18)
end for
obtain fusion result $\mathbf{H}_{\text{out}} = \tilde{\mathbf{H}}$
End

multiplication, and transforms result through linear projection to obtain intermediate $\mathbf{Z}_{\text{ms-ac}}^0$. Then, normalize output via Norm(·) layer and process through MLP layer to yield final $\mathbf{Z}_{\text{ms-ac}}$.

Similarly, the feature $\mathbf{Z}_{\text{ac-ms}}$ is generated through an identical computational pathway, with a critical role reversal. Input roles are swapped in the generation of $\mathbf{K}$, $\mathbf{Q}$, and $\mathbf{V}$. $\mathbf{Q}$ and $\mathbf{V}$ are derived from the accumulated matrix $\mathbf{Z}'$, while $\mathbf{K}'$ is generated jointly from $\mathbf{Z}'$ and multisource matrix $\mathbf{Z}^t$. This symmetric swap means (23) uses $\mathbf{Z}^t$ where $\mathbf{Z}'$ previously appeared, maintaining architectural symmetry while reversing feature dominance.

The multisource and accumulated joint feature $\mathbf{Z}_f$ is obtained by summing $\mathbf{Z}_{\text{ms-ac}}$ and $\mathbf{Z}_{\text{ac-ms}}$, followed by block-wise positional decoding. This matrix then undergoes sequential processing of convolution operation, positional encoding, feature normalization, positional decoding, and final convolution, to obtain the MCST global learning output $\mathbf{Z}_{\text{out}}$. The complete transformation is formally expressed as follows:

$$\mathbf{Z}_{\text{out}}^0 = \text{Norm}\left(\text{Encoding}\left(\text{Conv}\left(\mathbf{Z}_f\right)\right)\right) \tag{28}$$

$$\mathbf{Z}_{\text{out}} = \text{Conv}\left(\text{Decoding}\left(\mathbf{Z}_{\text{out}}^0\right)\right). \tag{29}$$

D. Cyclic-Refined Progressive Fusion

The CRPF module integrates local and global features from both branches, including spatial matrix $\mathbf{X}_{\text{BT}}^t$ and spectral matrix $\mathbf{X}_{\text{RT}}^t$ from PGCB and $\mathbf{Z}_{\text{out}}$ from MCST. These components undergo fusion through weighted summation scaled by parameter λ and scaling multiplication by coefficient Θ. Local and global features are integrated adaptively for the subsequent progressive fusion and result reconstruction. The resulting fused representation $\mathbf{H}$ serves as the CRPF module input, formally expressed as follows:

$$\mathbf{H} = \Theta \cdot \left(\mathbf{X}_{\text{BT}}^t + \mathbf{X}_{\text{RT}}^t + \lambda \cdot \mathbf{Z}_{\text{out}}\right). \tag{30}$$

Here, λ and Θ are automatically selected through iteration. The input matrix $\mathbf{H}$ first undergoes simple processing before nonlinear deep fusion, described as follows:

$$\mathbf{H}^0 = \text{ReLU}\left(\text{Conv}\left(\mathbf{H}\right)\right). \tag{31}$$

The feature map $\mathbf{H}^0$ is then processed through the progressive design, which is structured as a series of identical base blocks. The architecture of each block is fixed, and the only variable is the dilation rate within its dilated convolution layer, which increases progressively with network depth. To effectively process the input, which contains a hybrid of global and local representations, the fusion module should accommodate the distinct receptive fields these perspectives require. Consequently, we employ a strategy that progressively expands the receptive field, coupled with incremental feature fusion. This hierarchical process is formally defined by

$$\mathbf{H}' = \text{Dilated_Conv}\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{H}^0\right)\right), n\right) + \mathbf{H}^0. \tag{32}$$

The operator Dilated_Conv(·) denotes the dilated convolution, and n is the current layer index. As n increases, the receptive field of each layer dilates accordingly, using n as the direct scaling factor. This architecture facilitates comprehensive multiscale feature fusion and reconstruction, enabling a gradual and refined output generation process.

Finally, the HR-HSI is reconstructed based on the fused features using the following equation:

$$\tilde{\mathbf{H}} = \text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(\mathbf{H}'\right)\right)\right) + \hat{\mathbf{X}}_\mathbf{e}^\mathbf{t}. \tag{33}$$

This output $\tilde{\mathbf{H}}$ corresponds to a single processing cycle. The model executes multiple cycles per iteration, indexed by t. The output of cycle t, denoted $\tilde{\mathbf{H}}$, serves as the input $\hat{\mathbf{X}}_e^{t+1}$ for the subsequent cycle, feeding into both the MCST and PGCB branches for renewed feature extraction. Each cycle iteratively refines the features from the previous one, progressively enhancing the quality of feature extraction and fusion-based reconstruction. Residual learning and multicycle recursion are jointly employed to optimize performance. In each cycle, both $\hat{\mathbf{X}}_e^t$ and the multisource matrix $\mathbf{Z}^t$ are updated. This dynamic input modification optimizes cross-modal pairing, which enables the network to recover features potentially lost in earlier cycles and reassess complementary intermodal information.

The model processes LGCL and CRPF through multiple iterative cycles. It's a process of successive approximation, and the cyclic architecture explicitly models this intuition. The final output $\mathbf{H}_{\text{out}}$ from these cycles is compared to the ground-truth (GT) image to compute the reconstruction loss. Optimization is driven by the L_1 loss function, defined as follows:

$$Loss_{L_1} = \frac{1}{N_s}\sum_{i=1}^{N_s}\left|\text{GT}_i - \mathbf{H}_{\text{out}_i}\right| \tag{34}$$

where N_s represents the number of samples. The complete forward pass of LGP-Net, including the cyclic refinement process and variable updates across iterations, is summarized in Algorithm 1.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

IV. Experiments and Analysis

This section presents experimental validation and analysis. We benchmark against state-of-the-art methods, demonstrating fusion performance while validating framework efficacy. Ablation studies, parameter sensitivity analyses, and computational complexity assessments further substantiate our design.

A. Datasets

The CAVE dataset [46] comprises 32 images of indoor scenes, capturing the visible spectrum from 400 to 700 nm at a 10-nm spectral resolution (31 bands). Each image has a spatial resolution of 512 × 512 pixels and features a variety of objects, including toys, fruits, textiles, and food items.

The Harvard dataset [47] offers greater scene diversity with 50 images of both indoor and outdoor environments. It covers a broader visible range (420–720 nm) with the same spectral resolution (31 bands at 10 nm intervals). The original high-resolution images (approximately 1392 × 1040 pixels) are uniformly cropped to a standard size of 512 × 512 pixels.

The WDCM dataset [48] captures a broad spectral range from 400 to 2400 nm across 210 original bands. After the removal of noisy or water-absorption bands, 191 bands are retained for analysis. The image dimensions are 1280 × 296 pixels, encompassing diverse land cover categories such as buildings, grasslands, roads, and rivers.

The Chikusei dataset [49] covers the spectral range of 363–1018 nm across 128 bands at approximately 5 nm spectral resolution. The original high-resolution imagery (2517 × 2335 pixels) is uniformly cropped to a standardized size of 512 × 512 pixels for processing. The scene encompasses 19 distinct land cover classes, including farmland, buildings, roads, and others.

The YRE dataset [45] is a real-world full-resolution remote sensing dataset acquired over the eastern Yellow River Delta in China. The HSI is captured by the Gaofen-5 Advanced Hyperspectral Imager, and the MSI by the Sentinel-2A multispectral sensor. The HSI has a spatial size of 1400 × 1400 with 280 spectral bands (400–2500 nm), while the MSI has a spatial size of 4200 × 4200 with four bands (430–680 nm). Their spatial resolutions are 30 and 10 m, respectively.

The LN01 dataset [50] is collected over the northwest of Jinpu New District, Dalian City, China. After preprocessing, the HSI has a spatial size of 300 × 300 with 166 spectral bands, and the MSI has a spatial size of 900 × 900 with eight bands. The scene contains ten land-cover categories, including reservoir, seawater, sandy soil, broken bridge, barren grass, bare soil, highway, railway, mountain vegetation, and arable land.

B. Experimental Setup

In our implementation, the model optimization is performed using the Adam optimizer [51] for 500 epochs with a batch size of 8 and an initial learning rate of $2e^{-4}$. The proposed model is implemented using PyTorch 1.13.1 with CUDA 11.3. The hardware platform consists of an Intel Xeon Silver 4314 CPU (2.40 GHz, 32 physical cores) and an NVIDIA A40 GPU (48 GB memory). Detailed environment specifications are as follows: Python 3.9, CUDA driver version 11.4, and cuDNN 8.9. We benchmark our proposed method against eight state-of-the-art approaches. The comparison includes four CNN-based models: TF-Net [39], SSR-Net [24], DHIF [11], and MoG-DCN [10]; alongside four Transformer-based architectures: PSRT [42], 3DT [41], BDT [26], and DCT [13]. To ensure a fair comparison, all benchmark models are implemented using the hyperparameters specified in their original publications. All models are trained and evaluated on identical datasets under the same conditions to achieve their optimal performance.

TABLE I
AVERAGE PSNR, SSIM, SAM, RMSE, AND ERGAS OF DIFFERENT METHODS ON THE CAVE DATASET

Methods	CAVE ×8				
	PSNR	SSIM	SAM	RMSE	ERGAS
TF-Net	42.39	0.9902	4.3750	0.0078	1.8340
SSR-Net	45.49	0.9913	4.9230	0.0056	1.3920
DHIF	50.32	0.9955	2.8059	0.0031	0.7144
MoG-DCN	50.62	0.9954	2.7505	0.0030	0.6905
PSRT	50.12	0.9949	2.7266	0.0032	0.7252
3DT	50.37	0.9952	2.7601	0.0031	0.6961
BDT	50.81	0.9956	2.6810	0.0029	0.6685
DCT	50.66	0.9957	2.5243	0.0029	0.6478
LGP-Net	**51.49**	**0.9990**	**2.3031**	**0.0027**	**0.6070**
Ideal Value	∞	1	0	0	0

1) Data Processing: The observed LR-HSIs and HR-MSIs for all datasets (CAVE, Harvard, WDCM, and Chikusei) are simulated according to Wald's protocol [52]. The generation process for each HR-HSI involves applying a Gaussian blur followed by spatial downsampling to the GT image. The corresponding HR-MSI is synthesized by applying an SRF to the GT. For CAVE and Harvard datasets, the Nikon D700 SRF is used to generate three-band HR-MSI. For WDCM dataset, the Sentinel-2A SRF is used to generate ten-band HR-MSI. For Chikusei dataset, the published SRF is used to generate five-band HR-MSI. It is critical to note that the Gaussian filter and SRF are used for offline data preparation. To prevent data overlap between training and testing sets, a patch-based strategy with chosen dimensions for ×8 fusion is employed on each dataset.

2) Evaluation Metrics: The reconstruction quality is quantitatively evaluated using five standard metrics, including peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), root mean squared error (RMSE), spectral angle mapper (SAM), and relative dimensionless global error in synthesis (ERGAS). PSNR quantifies the ratio of maximum signal power to noise power. SSIM evaluates perceived quality based on luminance, contrast, and structure. RMSE measures average per-pixel error. SAM assesses spectral fidelity via average spectral angle. ERGAS combines spatial and spectral errors.

C. Comparison Experiments on Natural Images

1) Results of CAVE Dataset: The quantitative results of CAVE dataset are presented in Table I, where the best results are highlighted in bold, and the second-best results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

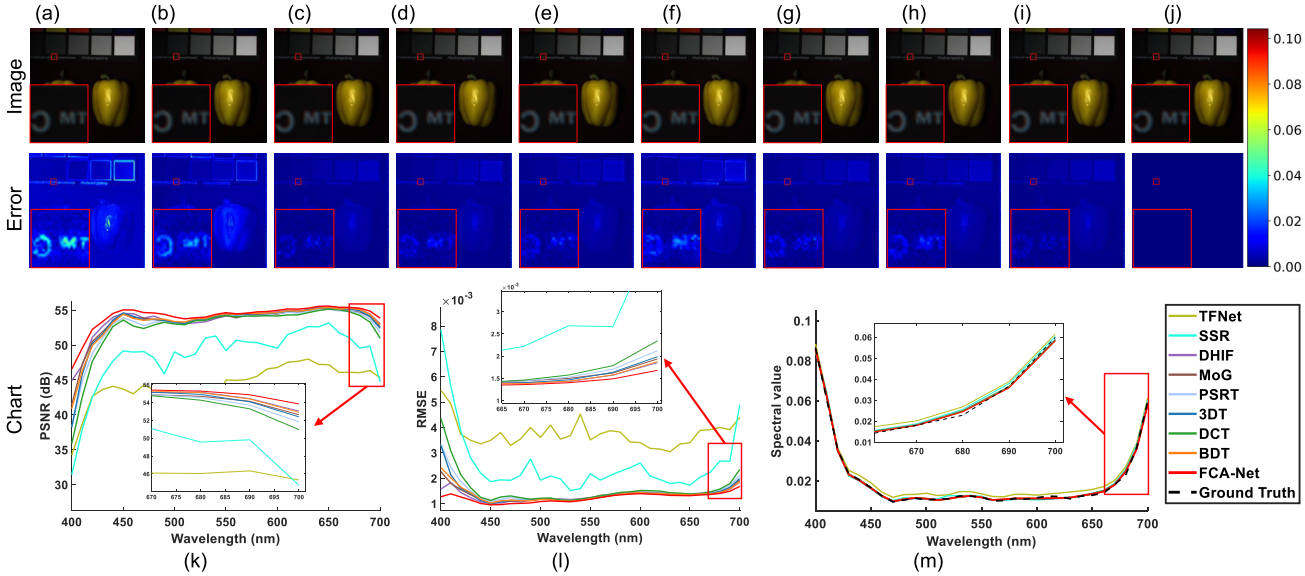


Fig. 4. Visualization comparison of different methods on the CAVE dataset. The above region shows the RGB color composites (R:29, G:19, B:9) of fused images and the corresponding error maps between each fused result and GT image for (a) TF-Net, (b) SSR-Net, (c) DHIF, (d) MoG-DCN, (e) PSRT, (f) 3DT, (g) BDT, (h) DCT, (i) LGP-Net, and (j) GT. The region below displays the quantitative and spectral comparisons for (k) PSNR of a test sample, (l) RMSE of a test sample, and (m) spectrum of a selected pixel.

are highlighted in underline. A qualitative comparison of the fusion outputs is provided in Fig. 4. Our proposed method outperforms other models across all evaluation metrics and demonstrates superior capabilities in spatial detail reconstruction and spectral fidelity preservation. To further validate these findings, we conduct a band-wise analysis on a representative test sample. As shown in Fig. 4(k) and (l), LGP-Net consistently achieves higher PSNR and lower RMSE across most spectral bands. The enlarged details focus on challenging wavelength intervals where curves converge in full-scale plots, revealing that our method maintains a clear advantage even in these difficult bands. This demonstrates that LGP-Net's superiority extends beyond average metrics to more stable reconstruction fidelity and lower per-band error along the spectral dimension. Furthermore, Fig. 4(m) compares spectral signatures at pixel (256, 256). The enlarged detail highlights critical regions of local peaks, valleys, and rapidly varying segments, where our method's curve aligns most closely with the GT. In contrast, competing methods exhibit varying degrees of spectral shift, amplitude deviation, or over-smoothing. This provides compelling evidence that LGP-Net preserves local spectral characteristics with superior fidelity.

2) Generalization Test on Harvard Dataset: To rigorously evaluate the model's generalization capability, we employ a cross-dataset validation strategy. Leveraging the identical spectral configuration of the CAVE and Harvard datasets (31-band HSI and 3-band MSI), the model is trained on the CAVE and evaluated on the Harvard. The quantitative results, presented in Table II, demonstrate that our model successfully generalizes to the unseen data domain. It achieves superior performance across most metrics, outperforming other state-of-the-art methods. Specifically, it attains a PSNR of 49.45 dB, surpassing the second-best method by a substantial margin, which indicates a remarkable improvement in reconstruction fidelity. SSIM values on the Harvard dataset are already near saturation (close to 1.0), making absolute differences among methods too small to be discriminative. Although LGP-Net does not achieve the highest SSIM, its value of 0.9810 still indicates strong structural fidelity. More importantly, SSIM mainly measures local structural similarity, while hyperspectral fusion demands a balance between spatial detail and spectral consistency. We posit that this is a consequential tradeoff, that our model prioritizes the preservation of spectral fidelity and avoids the spatial over-smoothing, which might artificially inflate SSIM. This focus is justified by the model's decisive improvements in metrics critically important to hyperspectral analysis, including SAM, RMSE, and ERGAS. The comprehensive superiority demonstrates that the slight compromise in spatial similarity is greatly outweighed by gains in overall reconstruction quality.

TABLE II

AVERAGE PSNR, SSIM, SAM, RMSE, AND ERGAS FOR DIFFERENT METHODS ON THE HARVARD DATASET

Methods	Harvard ×8				
	PSNR	SSIM	SAM	RMSE	ERGAS
TF-Net	41.81	0.9931	7.9233	0.0066	4.9911
SSR-Net	41.31	0.9885	11.0067	0.0068	8.4284
DHIF	46.13	0.9972	6.2320	0.0041	1.8902
MoG-DCN	45.51	0.9965	5.9466	0.0043	2.6858
PSRT	46.18	0.9971	6.4097	0.0041	2.1926
3DT	45.41	0.9968	6.3323	0.0043	2.3395
BDT	46.58	**0.9973**	5.5671	0.0038	2.0094
DCT	45.93	0.9971	6.0933	0.0041	2.0275
LGP-Net	**49.45**	0.9810	**5.5291**	**0.0033**	**1.6658**
Ideal Value	∞	1	0	0	0

D. Comparison Experiments on Remote Sensing Images

1) Results of WDCM Dataset: The qualitative fusion results of WDCM dataset are visualized in Fig. 5, with corresponding

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

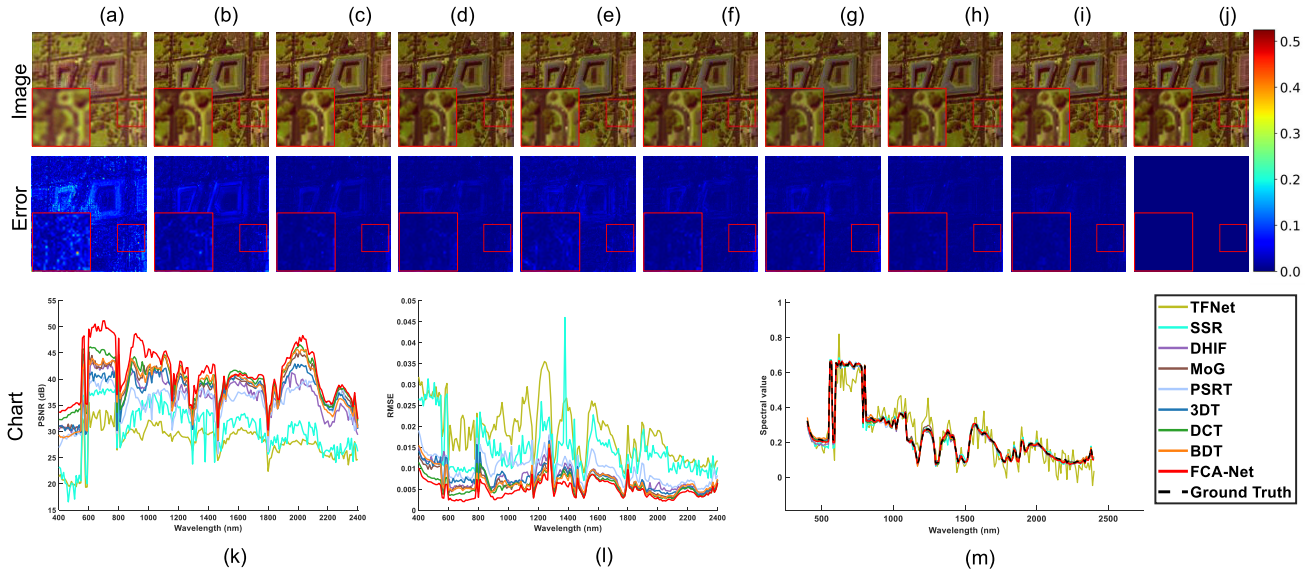


Fig. 5. Visualization comparison of different methods on the WDCM dataset. The above region shows the RGB color composites (R:55, G:120, B:7) of fused images and the corresponding error maps between each fused result and GT image for (a) TF-Net, (b) SSR-Net, (c) DHIF, (d) MoG-DCN, (e) PSRT, (f) 3DT, (g) BDT, (h) DCT, (i) LGP-Net, and (j) GT. The region below displays the quantitative and spectral comparisons for (k) PSNR of a test sample, (l) RMSE of a test sample, and (m) spectrum of a selected pixel.

TABLE III

AVERAGE PSNR, SSIM, SAM, RMSE, AND ERGAS FOR DIFFERENT METHODS ON THE WDCM DATASET

Methods	WDCM ×8				
	PSNR	SSIM	SAM	RMSE	ERGAS
TF-Net	33.92	0.9323	7.4920	0.0170	3.248
SSR-Net	36.11	0.9341	7.9501	0.0155	3.5686
DHIF	41.77	0.9829	5.3639	0.0079	1.6827
MoG-DCN	42.94	0.9886	2.5249	0.0069	1.5095
PSRT	40.12	0.9728	4.5345	0.0101	2.0545
3DT	42.25	0.9849	3.2457	0.0077	1.6506
BDT	43.30	0.9854	3.1011	0.0071	1.6546
DCT	44.26	0.9909	2.7317	0.0063	1.3136
LGP-Net	**45.27**	**0.9924**	**2.1806**	**0.0052**	**1.1592**
Ideal Value	∞	1	0	0	0

TABLE IV

AVERAGE PSNR, SSIM, SAM, RMSE, AND ERGAS FOR DIFFERENT METHODS ON THE CHIKUSEI DATASET

Methods	Chikusei ×8				
	PSNR	SSIM	SAM	RMSE	ERGAS
TF-Net	41.31	0.9716	4.1425	0.1547	2.1498
SSR-Net	41.82	0.9725	4.0669	0.1462	2.0369
DHIF	42.82	0.9767	3.5798	0.1122	1.7519
MoG-DCN	43.53	0.9779	3.5603	0.1035	1.7551
PSRT	43.73	0.9782	3.1702	0.1262	1.7005
3DT	43.78	0.9781	3.5971	0.1208	1.7357
BDT	44.47	0.9803	3.0534	0.0931	1.5968
DCT	43.37	0.9771	3.7860	0.1142	1.8042
LGP-Net	**44.96**	**0.9809**	**2.7443**	**0.0874**	**1.5394**
Ideal Value	∞	1	0	0	0

quantitative results provided in Table III. Our model demonstrates a significant performance improvement, achieving the highest scores across all evaluation metrics. Notably, it surpasses the second-best model (BDT) by nearly 1.0 dB in PSNR. This consistent dominance indicates superior overall fusion quality. To further validate these results, we conduct a granular analysis of spectral performance. The channel-wise PSNR and RMSE plots in Fig. 5 show that our model consistently outperforms all competitors across the entire spectral range, providing robust evidence of its robustness. Furthermore, the spectral signature at pixel (64, 64) is examined. While all methods avoid severe distortion, a detailed inspection reveals that our model's output most closely aligns with the GT curve, demonstrating its exceptional ability to preserve spectral fidelity.

2) Results of Chikusei Dataset: Qualitative results of Chikusei dataset are presented in Fig. 6. The corresponding quantitative evaluation is summarized in Table IV, which confirms that our model achieves state-of-the-art performance by outperforming all other methods across all metrics. Notably, it improves the PSNR by nearly 0.5 dB over the second-best model (BDT). A closer examination of the error maps in Fig. 6 reveals that our model produces fewer and less severe errors, particularly in regions where other models struggle. It effectively suppresses the formation of significant error spots, resulting in a more accurate reconstruction. This superiority is further substantiated by the channel-wise analysis in Fig. 6(k) and (l). The local magnification details demonstrate that our model consistently achieves higher PSNR and lower RMSE values across the entire spectral range, particularly in rapidly varying regions. Finally, the spectral fidelity of each method is compared at pixel (64, 64) in Fig. 6(m). Between 940 and 980 nm, other models exhibit noticeable spectral shifts, whereas the curve produced by LGP-Net aligns most closely with GT.

3) Results of Real-World YRE Dataset: To further evaluate the practical utility beyond simulated benchmarks, we conduct an additional experiment on the real-world YRE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

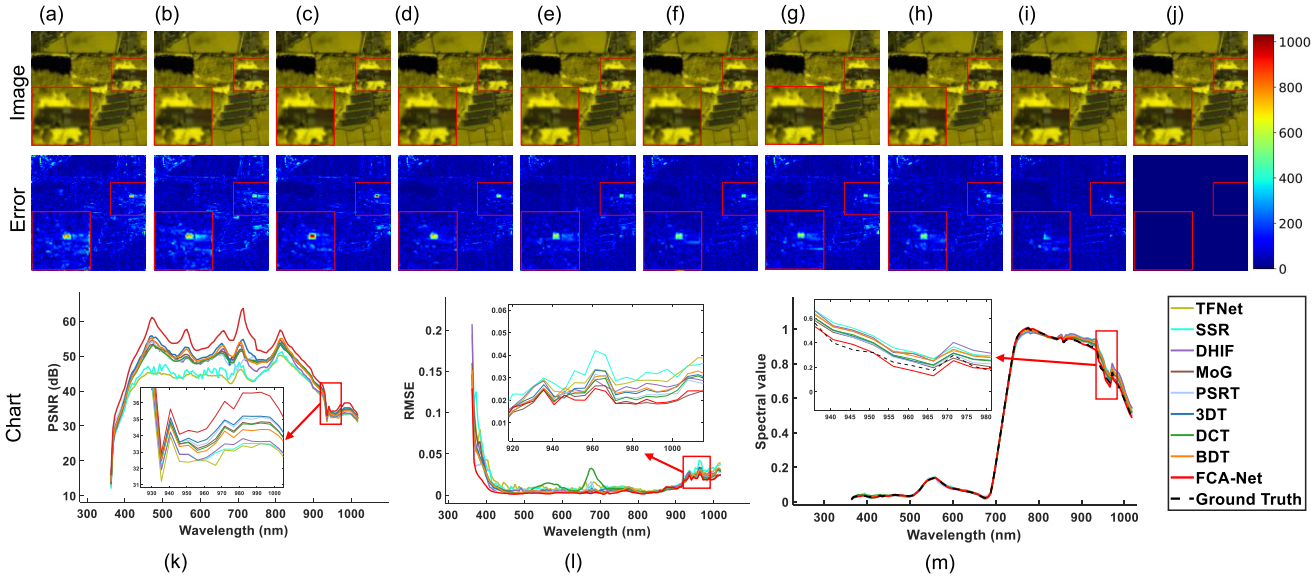


Fig. 6. Visualization comparison of different methods on the Chikusei dataset. The above region shows the RGB color composites (R:80, G:76, B:2) of fused images and the corresponding error maps between each fused result and GT image for (a) TF-Net, (b) SSR-Net, (c) DHIF, (d) MoG-DCN, (e) PSRT, (f) 3DT, (g) BDT, (h) DCT, (i) LGP-Net, and (j) GT. The region below displays the quantitative and spectral comparisons for (k) PSNR of a test sample, (l) RMSE of a test sample, and (m) spectrum of a selected pixel.

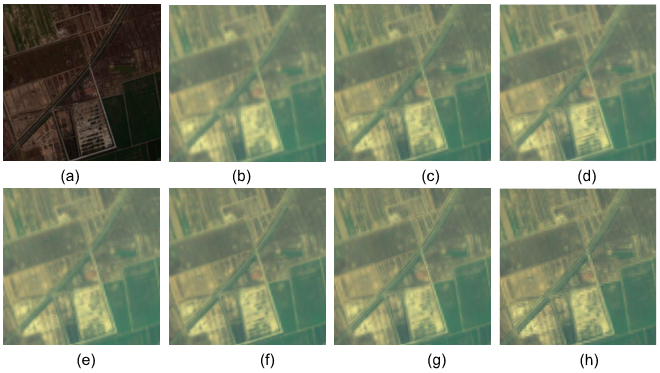


Fig. 7. Visual comparison on the real YRE dataset of (a) MSI (Sentinel-2), (b) HSI (Gaofen-5), (c) TF-Net, (d) SSR-Net, (e) MoG-DCN, (f) PSRT, (g) DCT, and (h) LGP-Net. The displayed fusion images are rendered as RGB composites using the 60th, 29th, and 7th spectral bands.

TABLE V
NO-REFERENCE QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE REAL YRE DATASET

Index	TF-Net	SSR-Net	MoG-DCN	PSRT	DCT	LGP-Net
QNR↑	0.8357	0.7735	0.7990	0.7874	0.8211	**0.8458**
D_λ ↓	0.0162	0.0194	0.0186	0.0184	0.0176	**0.0159**
D_s ↓	0.1678	0.2113	0.1858	0.1979	0.1797	**0.1507**

remote sensing dataset, where GT HR-HSI is unavailable. Full-resolution fusion is performed directly on real LR-HSI and HR-MSI observations. Following common practice in real-data HSI–MSI fusion, we adopt no-reference metrics for evaluation, including quality with no reference (QNR), D_λ, and D_s. A higher QNR indicates better overall fusion quality, while lower D_λ and D_s values indicate less spectral and spatial distortion, respectively. Table V reports the quantitative comparison. LGP-Net achieves the best performance among all compared

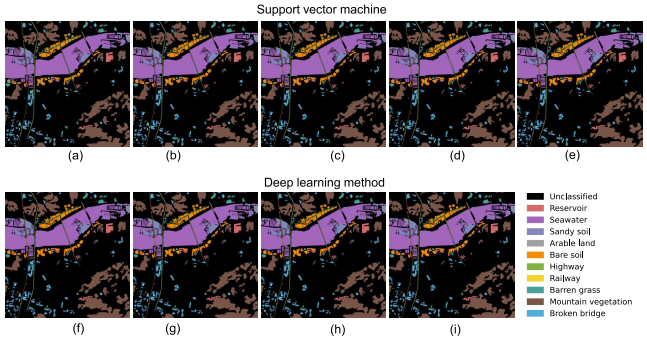


Fig. 8. Visual comparison of downstream land-cover classification results on the LN01 dataset. SVM for (a) HR-MSI, (b) LR-HSI upsampling, (c) DCT, (d) LGP-Net, and (e) GT, and deep learning for (f) HR-MSI, (g) LR-HSI upsampling, (h) DCT, and (i) LGP-Net.

methods, with the highest QNR of 0.8458 and the lowest D_λ and D_s values of 0.0159 and 0.1507, respectively. These results demonstrate that the proposed method preserves both spectral consistency and spatial structure more effectively than competing methods in a real-data setting without reference labels. Fig. 7 presents the visual comparison on the YRE dataset. Compared with other methods, LGP-Net produces clearer structural details and a more natural spatial–spectral appearance, with fewer visible artifacts and distortions. This real-data experiment further demonstrates that the proposed model generalizes well beyond simulated settings and is practically applicable to real-world HSI–MSI fusion tasks.

E. Downstream Land-Cover Classification Evaluation

Since a primary goal of HSI–MSI fusion is to provide high-quality data for practical remote sensing applications, we further evaluate the utility of the fused results in a downstream

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

QUANTITATIVE COMPARISON OF DOWNSTREAM LAND-COVER CLASSIFICATION PERFORMANCE ON THE LN01 DATASET USING HR-MSI, LR-HSI UPSAMPLING, AND FUSED RESULTS GENERATED BY DIFFERENT METHODS

Metrics	Support vector machine				Deep learning method			
	HRMSI	LRHSI-UP	DCT	LGP-Net	HRMSI	LRHSI-UP	DCT	LGP-Net
AA	91.73	93.52	95.66	**96.82**	98.68	98.41	98.36	**99.14**
OA	95.28	97.17	97.70	**98.01**	99.40	99.22	99.39	**99.51**
Kappa	93.42	96.05	96.79	**97.22**	99.16	98.91	99.15	**99.32**

TABLE VII

ABLATION EXPERIMENTAL RESULTS OF SSIF, LGCL, AND CRPF MODULES ON THE CAVE DATASET

SSIF	LGCL	CRPF	PSNR	SSIM	SAM	RMSE	ERGAS
✗	MCST	✗	49.89	0.9988	2.9312	0.0032	0.7386
✗	PGCB	✗	46.05	0.9974	3.3728	0.0051	1.1644
✗	✓	✗	50.14	0.9989	2.8671	0.0032	0.7233
✗	MCST	✓	50.95	0.9989	2.9162	0.0028	0.6441
✗	PGCB	✓	50.35	0.9988	3.1116	0.0031	0.6956
✗	✓	✓	51.17	0.9989	2.5532	0.0028	0.6383
✓	MCST	✗	49.99	0.9988	2.6615	0.0032	0.7428
✓	PGCB	✗	47.93	0.9981	3.3761	0.0041	0.9478
✓	✓	✗	51.04	0.9989	2.6893	0.0028	0.6482
✓	MCST	✓	50.96	0.9989	2.5466	0.0029	0.6529
✓	PGCB	✓	50.59	0.9988	2.6294	0.0030	0.6789
✓	✓	✓	**51.49**	**0.9990**	**2.3031**	**0.0027**	**0.6070**

TABLE VIII

ABLATION EXPERIMENTAL RESULTS OF SSIF, LGCL, AND CRPF MODULES ON THE WDCM DATASET

SSIF	LGCL	CRPF	PSNR	SSIM	SAM	RMSE	ERGAS
✗	MCST	✗	38.51	0.9895	5.3233	0.0116	2.6644
✗	PGCB	✗	31.18	0.9479	10.1917	0.0279	5.4826
✗	✓	✗	41.72	0.9951	4.9645	0.0081	1.7756
✗	MCST	✓	42.74	0.9961	4.4422	0.0071	1.5151
✗	PGCB	✓	39.18	0.9912	4.5166	0.0108	2.1838
✗	✓	✓	44.39	0.9973	2.6267	0.0058	1.2575
✓	MCST	✗	39.61	0.9919	4.5331	0.0102	2.5111
✓	PGCB	✗	33.30	0.9060	10.1575	0.0218	4.1858
✓	✓	✗	44.48	0.9913	2.2827	0.0058	1.3372
✓	MCST	✓	43.35	0.9965	3.7803	0.0066	1.4365
✓	PGCB	✓	39.91	0.9924	4.3833	0.0098	2.1842
✓	✓	✓	**45.27**	**0.9924**	**2.1806**	**0.0052**	**1.1592**

land-cover classification task. To this end, we conduct classification experiments on the LN01 dataset, which contains ten terrain categories. We compare classification performance on four input types, including HR-MSI, LR-HSI upsampling, the fused result of DCT (stable comparison method), and the fused result of LGP-Net. For comprehensive evaluation, both a traditional support vector machine (SVM) classifier and a deep learning-based classifier [53] are adopted. Performance is evaluated using average accuracy (AA), overall accuracy (OA), and Kappa coefficient.

Table VI summarizes the quantitative results. The fused result generated by LGP-Net achieves the best performance under both classifier settings. With SVM, LGP-Net achieves AA/OA/Kappa values of 96.82/98.01/97.22, outperforming the HR-MSI, LR-HSI, and DCT-based fusion results. With the deep learning-based classifier, the proposed method also achieves the best results, with AA/OA/Kappa values of 99.14/99.51/99.32, respectively. These improvements indicate that the fused data produced by LGP-Net contain more discriminative spatial–spectral information for classification. The visual classification maps in Fig. 8 further support this conclusion. Compared with other inputs, the classification result based on LGP-Net exhibits clearer class boundaries, fewer misclassified regions, and better preservation of fine spatial structures. Therefore, the proposed method not only improves reconstruction fidelity but also provides more effective input data for downstream remote sensing applications.

F. Ablation Analysis

1) SSIF Module: The SSIF module acts as a front-end preprocessing unit, designed to refine the input representation prior to deep feature extraction, which is demonstrated through ablation experiments in Tables VII–IX. The results of WDCM dataset (Table VIII) demonstrate the SSIF's consistent benefits. When added to the MCST, the SSIF yields a marginal PSNR improvement from 38.51 to 39.61 dB and an SSIM gain from 0.9895 to 0.9919, accompanied by a reduction in SAM. This indicates that while the SSIF alone does not produce a dramatic performance shift, it provides a stable foundation and a beneficial regularization that facilitates downstream processing. The value of the SSIF becomes more pronounced in a complete system. When integrated with both the PGCB and MCST, performance improves markedly, with PSNR increasing from 41.72 to 44.48 dB, and SAM decreasing significantly from 4.9645 to 2.2827. This same trend is consistently observed across the CAVE and Chikusei datasets

TABLE IX

ABLATION EXPERIMENTAL RESULTS OF SSIF, LGCL, AND CRPF MODULES ON THE CHIKUSEI DATASET

SSIF	LGCL	CRPF	PSNR	SSIM	SAM	RMSE	ERGAS
✗	MCST	✗	43.65	0.9716	3.6145	0.1023	1.7664
✗	PGCB	✗	39.99	0.9486	4.2392	0.1593	2.2306
✗	✓	✗	43.81	0.9718	3.6083	0.1002	1.7357
✗	MCST	✓	44.18	0.9731	3.0916	0.0955	1.6276
✗	PGCB	✓	44.36	0.9735	3.2331	0.0942	1.6137
✗	✓	✓	43.56	0.9712	3.5557	0.1002	1.7071
✓	MCST	✗	43.79	0.9718	3.3861	0.1001	1.7134
✓	PGCB	✗	41.34	0.9615	4.1056	0.1192	2.0066
✓	✓	✗	44.33	0.9735	3.2384	0.0944	1.6371
✓	MCST	✓	44.75	0.9746	2.7847	0.0897	1.5557
✓	PGCB	✓	44.71	0.9744	2.9457	0.0901	1.5732
✓	✓	✓	**44.96**	**0.9809**	**2.7443**	**0.0874**	**1.5394**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

TABLE X

COMPARISON OF RECONSTRUCTION PERFORMANCE FOR DIFFERENT ATTENTION SUBMODULES ON THE WDCM DATASET

Methods	PSNR	SSIM	SAM	RMSE	ERGAS
BDT	37.37	0.9512	6.3982	0.0132	2.8073
DCT	40.57	0.9811	3.3210	0.0092	1.9961
MCST	**45.27**	**0.9924**	**2.1806**	**0.0052**	**1.1592**

(Tables VII and IX). These findings confirm that the SSIF module is not merely a passive preprocessor but a critical foundational component, establishing robust representation.

2) LGCL Module: To evaluate the effectiveness of feature extraction components, we ablate the PGCB and MCST submodules both separately and jointly. In Table VIII, when the MCST is used in isolation, it achieves a PSNR of 38.51 dB and an SSIM of 0.9895. In contrast, the PGCB alone yields inferior results (PSNR: 31.18 dB, SSIM: 0.9479). This stark performance gap underscores that the MCST provides significantly more robust spatial–spectral representations than the PGCB when operating independently. Furthermore, their combination produces a synergistic effect, boosting performance to 41.72 dB PSNR and 0.9951 SSIM. This demonstrates that the PGCB and MCST offer complementary strengths, with the former capturing multiscale local features and the latter modeling global contextual dependencies. This synergy is further amplified when integrated with the CRPF fusion module, achieving 44.39 dB PSNR and 0.9973 SSIM. This confirms that sophisticated feature extraction must be coupled with advanced fusion mechanisms to maximize model capacity. These performance trends are consistently observed in Tables VII and IX, validating the generalizability of module interaction.

To further validate the MCST's design, we replaced it with attention submodules from state-of-the-art models (DCT and BDT) while keeping all other components unchanged. As shown in Table X, our MCST module outperforms these alternatives, demonstrating its superior competitiveness and contribution to the overall model performance. The MCST module extracts global spectral–spatial features through CCA with cyclic accumulation features.

3) CRPF Module: The CRPF module is designed to integrate multiscale PGCB and MCST features, and its inclusion consistently enhances performance across all metrics in ablation results. For example, in Table VIII, adding the CRPF module to the standalone MCST increases the PSNR from 38.51 to 42.74 dB and reduces the SAM from 5.3233 to 4.4422. More strikingly, when integrated with the PGCB, the CRPF facilitates a substantial performance recovery, elevating the PSNR from 31.18 to 39.18 dB. These results demonstrate that the CRPF not only effectively consolidates feature representations but also compensates for the limitations of individual components. The optimal performance is achieved in the configuration that integrates SSIF, PGCB + MCST, and CRPF modules. A comparison with the configuration that lacks the CRPF reveals measurable gains attributable solely to the CRPF, underscoring its critical role as the ultimate integrator of high-level features. These performance trends are consistently observed in Tables VII and IX.

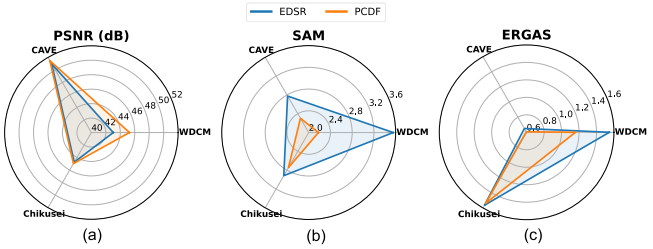


Fig. 9. Ablation experimental results of the CRPF module. Three graphs represent (a) PSNR, (b) SAM, and (c) ERGAS scores of the model on CAVE, WDCM, and Chikusei datasets.

TABLE XI

IMPACT OF DIFFERENT CYCLIC ACCUMULATION STRATEGIES ON HR-HSI RECONSTRUCTION ON THE WDCM DATASET

Methods	PSNR	SSIM	SAM	RMSE	ERGAS
DCT	40.57	0.9811	3.3210	0.0092	1.9961
MCST-	40.87	0.9818	3.7964	0.0089	1.9398
DCT+	43.17	0.9872	4.6830	0.0069	1.5003
MCST	**45.27**	**0.9924**	**2.1806**	**0.0052**	**1.1592**

To further validate the CRPF, we conduct an experiment replacing it with the widely recognized enhanced deep super-resolution (EDSR) module [54], a strong baseline for feature fusion. As illustrated in Fig. 9, under identical experimental conditions, our CRPF module consistently outperforms the EDSR-based variant, achieving higher PSNR and lower SAM and ERGAS values. These findings confirm that the CRPF module is not only a substitute for existing fusion blocks but also a superior reconstruction design.

4) Cyclic Accumulation Strategy: To validate the efficacy of the cyclic accumulation strategy within the MCST submodule, we conduct a dedicated ablation study, with results detailed in Table XI. The experiment compares three configurations: DCT replaces the MCST with a core attention module from the DCT model, DCT+ enhances the DCT module by integrating our cyclic accumulation strategy, and MCST–means the MCST module without the cyclic accumulation strategy. The quantitative results demonstrate a consistent trend that the incorporation of a cyclic accumulation strategy yields performance improvements across the board. Notably, both the external DCT module and our MCST architecture achieve higher metrics when equipped with this strategy. This provides clear empirical evidence that the cyclic accumulation mechanism is a significant contributor to enhanced model performance, independent of the specific attention module it augments.

G. Parameter Analysis

1) Number of Iterative Cycles: During our model's development, we observed that performance is improved with an increased number of iterative cycles. This effect is further amplified by the integration of the cyclic accumulation strategy within the MCST submodule. To quantitatively determine the optimal number of cycles, we conduct a parameter study, varying the cycle count T from 1 to 6. The results, shown in Fig. 10(a)–(c), demonstrate that performance peaks at $T = 4$. We posit that this optimum represents a balance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

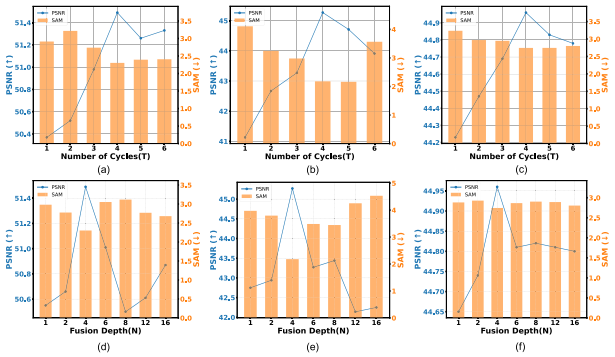


Fig. 10. Parameter analysis of the cycle number T and the fusion depth N on three datasets: (a) CAVE-T, (b) WDCM-T, (c) Chikusei-T, (d) CAVE-N, (e) WDCM-N, and (f) Chikusei-N.

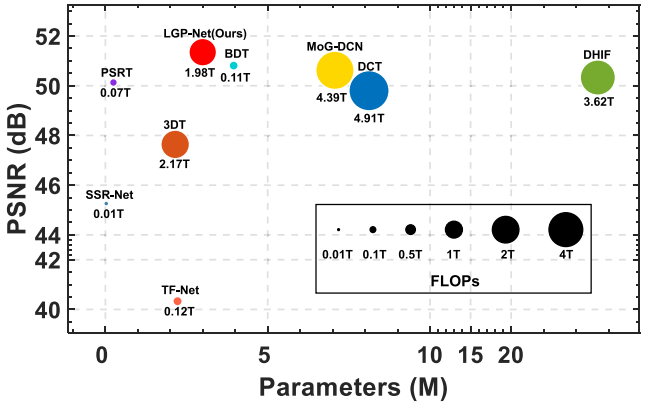


Fig. 11. Performance comparison of various methods. Horizontal axis indicates model parameters (M), vertical axis denotes PSNR (dB), and circle size represents trillions of floating-point operations (T).

where the accumulated features contain rich information while minimizing mutual interference, thereby providing the most effective representation for the fusion process. Performance improves steadily with increasing cycles for $T < 4$, which indicates that iterative cycling effectively refines features up to an optimal point. However, beyond $T = 4$, both PSNR and SAM slightly degrade, suggesting that excessive cycles introduce feature redundancy and noise accumulation.

2) Fusion Depth of CRPF Module: The fusion depth N of the CRPF module is a critical hyperparameter that significantly influences reconstruction quality. To quantitatively evaluate its impact, we conduct a series of parameter experiments under identical conditions, varying the depth N from 1 to 16 while performing fusion-based reconstruction. As illustrated in Fig. 10(d)–(f), the optimal performance is achieved at a fusion depth of $N = 4$. This suggests that a four-layer configuration provides the optimal receptive field size and representational capacity for effectively integrating multiscale features, thereby maximizing the quality of the fused output. Performance improves as the depth N increases from shallow to moderate, which reflects progressively stronger spatial–spectral integration. However, beyond $N = 4$, performance becomes unstable, suggesting that excessive depth introduces optimization difficulties and potential overfitting.

H. Model Complexity Analysis

This section analyzes the computational complexity and performance of various models on the CAVE dataset, as summarized in Fig. 11. Our LGP-Net model achieves the highest PSNR while maintaining parameter counts and FLOPs within the top five most efficient, indicating a favorable balance between performance and complexity. This efficiency stems from a design that synergistically combines global contextual modeling via attention with local feature extraction via convolution, resulting in complementary representations that drive superior reconstruction. Further analysis of Fig. 11 shows that some high-accuracy models (e.g., DCT, MoG-DCN) incur substantial computational costs, while lightweight alternatives (e.g., PSRT, SSR-Net) suffer from significant performance degradation. In contrast, our model occupies an optimal position in this design space, delivering state-of-the-art accuracy with moderate computational demands. This balance between high fidelity and practical efficiency makes it particularly suitable for resource-constrained or real-world application scenarios.

V. Conclusion

This article presents an LGP-Net for hyperspectral and multispectral image fusion. The proposed framework begins with an SSIF module for input preprocessing and preliminary reconstruction. The output is then fed into an LGCL module for feature extraction, including PGCB for local details and MCST for global dependencies. A cyclic accumulation strategy within the MCST further enhances its representational ability. The local and global features are integrated by a CRPF module for expanding receptive fields and reconstructing HR-HSI. Extensive experiments on multiple benchmarks demonstrate that LGP-Net consistently outperforms state-of-the-art methods in both spatial and spectral fidelity. Although LGP-Net demonstrates strong spectral fidelity overall, we observe that in spectral regions with extremely narrow absorption features, the reconstruction error can be slightly higher than the average. This is because such sharp spectral variations are inherently more difficult to reconstruct from downsampled or multispectral observations. Future work could explore incorporating explicit spectral priors or physics-based constraints to better capture these fine spectral details.

References

[1] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.

[2] X. Fu, X. Zhou, Y. Fu, P. Liu, and S. Jia, "Progressive semantic enhancement network for hyperspectral and LiDAR classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 6, pp. 11327–11339, Jun. 2025.

[3] X. Zhang et al., "TFDet: Target-aware fusion for RGB-T pedestrian detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 7, pp. 13276–13290, Jul. 2025.

[4] B. Tu, W. He, Q. Li, Y. Peng, and A. Plaza, "A new context-aware framework for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505114.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

[5] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[6] C. Li, B. Zhang, D. Hong, X. Jia, A. Plaza, and J. Chanussot, "Learning disentangled priors for hyperspectral anomaly detection: A coupling model-driven and data-driven paradigm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6883–6896, Apr. 2025.

[7] W. Wang, W. Zeng, Y. Huang, X. Ding, and J. Paisley, "Deep blind hyperspectral image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4150–4159.

[8] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023.

[9] N. Wang, A. Guo, R. Dian, and S. Li, "Equivariant high-resolution hyperspectral imaging via mosaiced and PAN image fusion," *IEEE Trans. Image Process.*, vol. 35, pp. 1246–1260, 2026.

[10] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[11] T. Huang, W. Dong, J. Wu, L. Li, X. Li, and G. Shi, "Deep hyperspectral image fusion network with iterative spatio-spectral regularization," *IEEE Trans. Comput. Imag.*, vol. 8, pp. 201–214, 2022.

[12] J. Li, K. Zheng, J. Yao, L. Gao, and D. Hong, "Deep unsupervised blind hyperspectral and multispectral data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Reciprocal transformer for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102148.

[14] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[15] H. Zhang, H. Xu, X. Tian, J. Jiang, and J. Ma, "Image fusion meets deep learning: A survey and perspective," *Inf. Fusion*, vol. 76, pp. 323–336, Dec. 2021.

[16] Z. Shi, C. Chen, Z. Xiong, D. Liu, Z.-J. Zha, and F. Wu, "Deep residual attention network for spectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2019, pp. 214–229.

[17] X. Deng and P. L. Dragotti, "Deep convolutional neural network for multi-modal image restoration and fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3333–3348, Oct. 2021.

[18] Q. Xu, S. Liu, J. Wang, B. Jiang, and J. Tang, "AS3ITransUNet: Spatial–spectral interactive transformer U-net with alternating sampling for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5523913.

[19] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[20] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, 2020, pp. 6840–6851.

[21] R. Ran, L.-J. Deng, T.-X. Jiang, J.-F. Hu, J. Chanussot, and G. Vivone, "GuidedNet: A general CNN fusion framework via high-resolution guidance for hyperspectral image super-resolution," *IEEE Trans. Cybern.*, vol. 53, no. 7, pp. 4148–4161, Jul. 2023.

[22] C. Zhu, S. Deng, Y. Zhou, L.-J. Deng, and Q. Wu, "QIS-GAN: A lightweight adversarial network with quadtree implicit sampling for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531115.

[23] J. Qu, J. He, W. Dong, and J. Zhao, "S2CycleDiff: Spatial–spectral-bilateral cycle-diffusion framework for hyperspectral image super-resolution," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 5, pp. 4623–4631.

[24] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

[25] J.-F. Hu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, D. Hong, and G. Vivone, "Fusformer: A transformer-based fusion network for hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] S. Deng, L.-J. Deng, X. Wu, R. Ran, and R. Wen, "Bidirectional dilation transformer for multispectral and hyperspectral image fusion," in *Proc. 32nd Int. Joint Conf. Artif. Intell.*, Aug. 2023, pp. 3633–3641.

[27] W. Xie, Y. Cui, Y. Li, J. Lei, Q. Du, and J. Li, "HPGAN: Hyperspectral pansharpening using 3-D generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 463–477, Jan. 2021.

[28] J. Yin, J. Qu, Q. Chen, M. Ju, and J. Yu, "Differential strategy-based multi-level dense network for pansharpening," *Remote Sens.*, vol. 14, no. 10, p. 2347, May 2022.

[29] C. Zhu, S. Deng, X. Song, Y. Li, and Q. Wang, "Mamba collaborative implicit neural representation for hyperspectral and multispectral remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504915.

[30] W. He, Y. Cai, Q. Ren, A. Ruze, and S. Jia, "Adaptive expert learning for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5528615.

[31] Y. Su et al., "Dilated transformation-guided unsupervised multimodal learning for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5535715.

[32] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[33] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Apr. 2023, Art. no. 5509417.

[34] J. Xu et al., "FusGAT: Graph attention-based fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[35] K. Zheng, A. Chen, H. Yu, J. Li, and B. Zhang, "Unsupervised pretraining framework guided hyperspectral and multispectral image fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[36] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Multispectral and hyperspectral image fusion using a 3-D-convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 639–643, May 2017.

[37] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Pyramid fully convolutional network for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 5, pp. 1549–1558, May 2019.

[38] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

[39] X. Liu, Q. Liu, and Y. Wang, "Remote sensing image fusion based on two-stream fusion network," *Inf. Fusion*, vol. 55, pp. 1–15, Mar. 2019.

[40] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C. Chan, and W. Philips, "Spatial–spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[41] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Learning a 3D-CNN and transformer prior for hyperspectral image super-resolution," *Inf. Fusion*, vol. 100, Dec. 2023, Art. no. 101907.

[42] S.-Q. Deng, L.-J. Deng, X. Wu, R. Ran, D. Hong, and G. Vivone, "PSRT: Pyramid shuffle-and-reshuffle transformer for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503715.

[43] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.

[44] X. Wang, X. Wang, R. Song, X. Zhao, and K. Zhao, "MCT-Net: Multi-hierarchical cross transformer for hyperspectral and multispectral image fusion," *Knowl.-Based Syst.*, vol. 264, Mar. 2023, Art. no. 110362.

[45] W. He, X. Fu, N. Li, Q. Ren, and S. Jia, "LGCT: Local–global collaborative transformer for fusion of hyperspectral and multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537114.

[46] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[47] A. Chakrabarti and T. Zickler, "Statistics of real-world hyperspectral images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 193–200.

[48] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[49] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Dept. Space Applications Laboratory, University of Tokyo, Bunkyō, Japan, Tech. Rep. SAL-2016-05-27, 2016.

[50] H. Yu, Z. Ling, K. Zheng, L. Gao, J. Li, and J. Chanussot, "Unsupervised hyperspectral and multispectral image fusion with deep spectral–spatial collaborative constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534114.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.

[51] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[52] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogrammetric Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.
[53] M. Li, Y. Liu, G. Xue, Y. Huang, and G. Yang, "Exploring the relationship between center and neighborhoods: Central vector oriented self-similarity network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1979–1993, Apr. 2023.
[54] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 136–144.



Shuyu Zhang (Member, IEEE) received the B.S. and Ph.D. degrees from the College of Earth Sciences, Zhejiang University, Hangzhou, China, in 2015 and 2020, respectively.

She is currently an Assistant Professor with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. Her research interests include remote sensing image analysis, ocean data processing, and deep learning.



Yixun Cai received the B.E. degree in computer science and technology from Shenyang Jianzhu University, Shenyang, China, in 2023. He is currently pursuing the M.S. degree in computer science and technology with Shenzhen University, Shenzhen, China.

His research interests include hyperspectral image (HSI) fusion, super-resolution, and deep learning.



Wangquan He (Student Member, IEEE) received the M.S. degree in information and communication engineering from Hunan Institute of Science and Technology, Yueyang, China, in 2022. He is currently pursuing the Ph.D. degree with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China.

His research interests include hyperspectral image (HSI) processing and super-resolution.



Sen Jia (Senior Member, IEEE) received the B.E. and Ph.D. degrees from the College of Computer Science, Zhejiang University, Hangzhou, China, in 2002 and 2007, respectively.

Since 2008, he has been with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China, where he is currently a Full Professor. His research interests include remote sensing image processing, signal and image processing, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:15:45 UTC from IEEE Xplore. Restrictions apply.